ADC (Automated Dispensing Cabinet) - System Guide

The ADC is the cabinet side of the project. It is written in Python 3.10+ with the standard library only. The physical cabinet is simulated by a local web page (the "panel"). Logins are checked by the HMS (`monitor/`) over TCP.

1. How to run it

From the repo root (use `py` on Windows, `python3` on macOS):

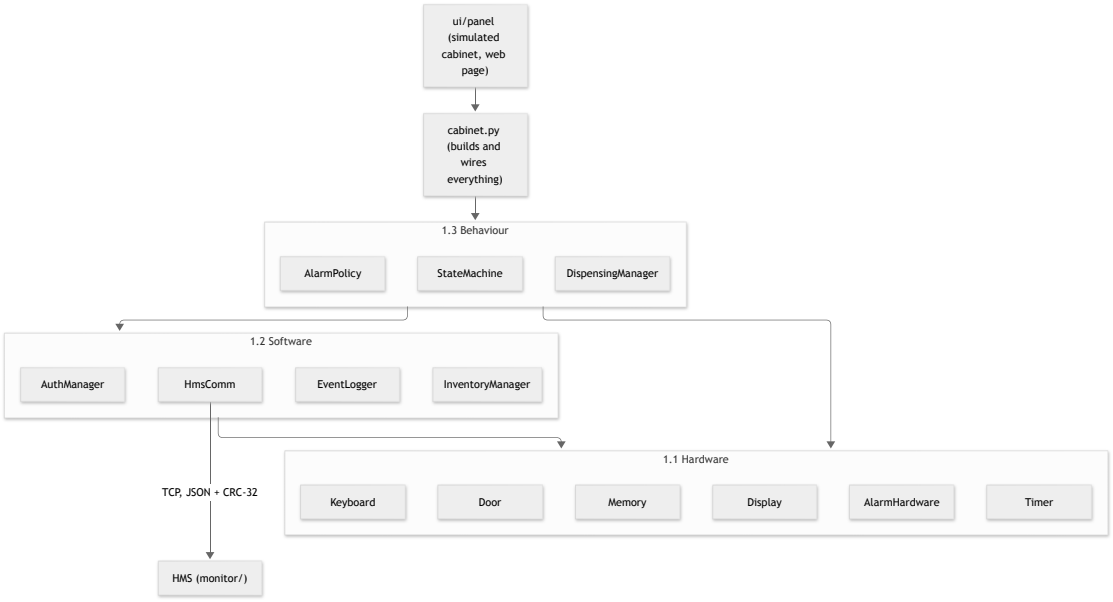
1. Start the HMS: `python3 monitor/src/main.py --sim-host` (HMS page: http://127.0.0.1:8080)
2. Start the cabinet: `python3 adc/src/adc_main.py` (panel: http://127.0.0.1:8090)
3. Run the tests: `python3 -m unittest discover -s adc/tests` (20 tests, about 2 s)

Each demo password is the user name followed by `!demo`, for example `nurse1` / `nurse1!demo`.

Account	Role	What it can do at the cabinet
doctor1, nurse1	Doctor, Nurse	Dispense
pharmacist1	Pharmacist	Deposit (stock an order)
admin1, tech1	Administrator, Technician	Reset a lockout, disarm an alarm, maintenance mode (they cannot open the cabinet)

2. Architecture (the Deliverable 2 module guide)

The code follows the three layers of the module guide. Each layer only uses the layers below it.



3. Folder and file guide

```
adc/
├── ADC_GUIDE.pdf          this document
├── src/
│   ├── README.md          run instructions, module and requirements tables, design notes
│   ├── adc_main.py        entry point: starts one cabinet and its web panel
│   ├── data/<id>/         the cabinet's saved memory (created at run time, not in git)
│   └── adc/               the cabinet package
│       ├── cabinet.py
│       ├── hardware/      1.1 Hardware hiding
│       ├── software/      1.2 Software hiding
│       ├── behaviour/     1.3 Behaviour
│       └── ui/            the simulated cabinet (web panel)
└── tests/
    ├── README.md          what the tests cover
    └── test_adc.py        20 automated tests
```

Top level

File / folder	What it does
`src/adc_main.py`	Entry point. Parses `--id`, `--hms-host`, `--port`, `--capacity`, `--ui-port`, `--data` and `--lan`, powers on a `Cabinet`, and starts the `Panel`. It isn't called `main.py` because the ADC tests also import the HMS's `main.py`.
`src/data/<id>/`	Saved state that survives a power cut, as JSON files: `auth.json` (failed-login count and lockout flag), `inventory.json` (orders and compartments), `activity_log.json` (the event log) and `outbox.json` (messages waiting for the HMS). Delete the folder to start fresh.
`src/adc/__init__.py`	Package description: module numbers follow the Deliverable 2 "Modular Design and Interfaces" table.
`src/adc/cabinet.py`	`Cabinet`, module 1 (the entire system). Builds every module in layer order and connects them. Handles power on/off (the door stays latched on power loss), runs the 50 ms tick that drives the timers, and gives the panel its physical-world controls (`press`, `set_door_open`, `set_tamper`, `set_network`) and a status snapshot.

`hardware/` - 1.1 Hardware hiding (the physical devices)

File	Class	What it hides / does
`alarm_hw.py`	`AlarmHardware`	The siren. `alarmOn` / `alarmOff`, and suppression during maintenance.
`keyboard.py`	`Keyboard`, `InputEvent`	All user input (keypad, badge, barcode scanner, compartment sensor). Every input reaches the cabinet as one `InputEvent`, interrupt style.
`display.py`	`Display`	The cabinet screen. All on-screen wording lives in `MESSAGES`, and callers name a message by its ID. Also handles blocking messages, which must be dismissed, or clear when the cabinet leaves Invalid or Locked Out.
`door.py`	`Door`	Latch, door sensor and tamper switch. Reports "opened", "closed" and "tamper" events. A latched door refuses to open.
`timer.py`	`Timer`, `SimulatedTimer`	The clock. Other modules start and stop timings here and never read the clock directly, so tests can fast-forward time with `SimulatedTimer`.

File	Class	What it hides / does
`memory.py`	`Memory`	Storage that survives power loss: one JSON file per record, written atomically.

`software/` - 1.2 Software hiding (the rules and data)

File	Class	What it hides / does	Requirements
`auth.py`	`AuthManager`	Asks the HMS to check the password and counts consecutive failures: a warning on the 3rd, lockout on the 5th. Owns the 5-minute session and the lockout flag (both survive power loss). `verifyService` checks administrators and technicians for reset, disarm and maintenance.	FR-SEC-01/02/03/04/07, NFR-SEC-01/05
`inventory.py`	`InventoryManager`	Every order and its compartment. Order status: Pending, Stocked, Fulfilled or Cancelled. Deposit rules (wrong ADC, not on the order list, already stocked, cabinet full), the pharmacist's quantity confirmation, syncing with the HMS order list, and the `EVENT` / `INVENTORY` reports, including the 95% capacity warning.	FR-IVT-01 to 05, FR-HMS-02/04/06
`event_logger.py`	`EventLogger`	Time-stamped log of every action (who, what, order, medication, quantity, alarms with time to disarm and notes). `flushLog` uploads unsent entries to the HMS as `LOG_BATCH`.	FR-HMS-01/05
`hms_comm.py`	`HmsComm`	The TCP link to the HMS: one JSON frame per line with a CRC-32 checksum. Handles the heartbeat every 0.5 s, password checks answered within 500 ms, acknowledgements, and an offline outbox saved to memory and resent on reconnect. It follows the HMS clock, because the HMS rejects frames more than 60 s off.	FR-HMS-02/03/04/07, NFR-HMS-01, NFR-SEC-05

`behaviour/` - 1.3 Behaviour (what the cabinet does)

File	Class	What it does	Requirements
`state_machine.py`	`StateMachine`	The only place that decides the mode (see the diagram below). It locks and unlocks the door for each mode and checks the timed rules: 10 s relock if the door isn't opened, Invalid plus an alarm if the door is open more than 60 s (closing the door returns to Authorized), and logout after 5 minutes idle. The tamper switch sends it to Locked Out.	FR-SEC-02/05/08/09/10

File	Class	What it does	Requirements
`dispensing.py`	`DispensingManager`	The user workflow. It handles every keypad input (login, show order, dispense, scan deposit, confirm quantity, compartment done, cancel, dismiss, reset, disarm, maintenance), shows an order before dispensing, and runs the open-door, take/place, close-door sequence. It sends `ORDER_STATUS` (Stocked / Fulfilled) to the HMS, and explains why a latched door won't open.	FR-DSP-01/02, FR-HMS-03
`alarm_policy.py`	`AlarmPolicy`	Which conditions sound the alarm: the tamper switch (cabinet forced open or damaged) and the door open too long. It sends `ALARM` / `ALARM_CLEARED` to the HMS and logs each disarm with the time to disarm and notes. The door-open alarm stops when the door is closed. The tamper alarm keeps sounding until an administrator or technician disarms it.	FR-SEC-06, FR-HMS-05

`ui/` - the simulated cabinet

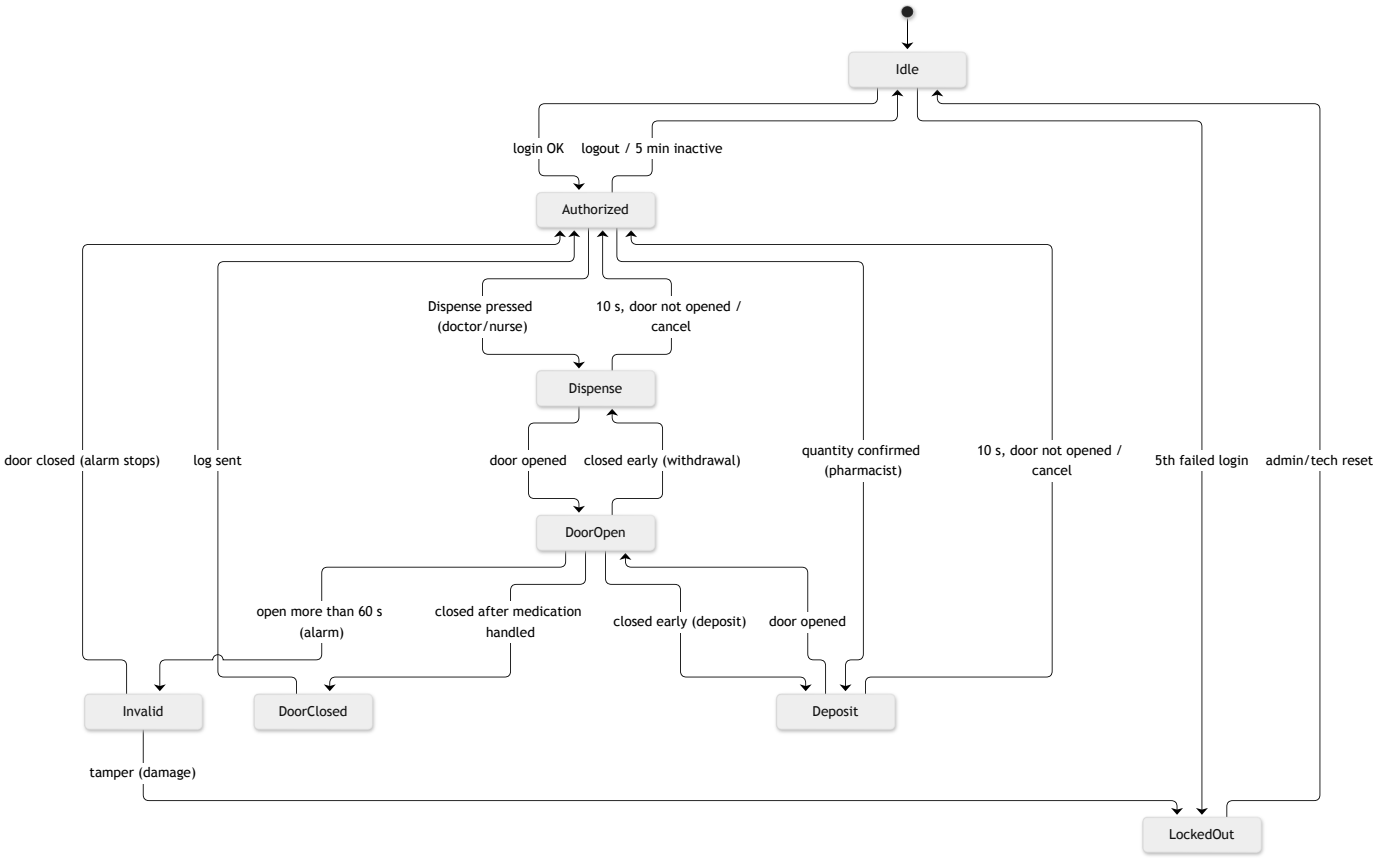
File	What it does
`panel.py`	`Panel`, a small local web server. `GET /api/state` returns the cabinet snapshot. `POST /api/input` (keypad), `/api/door`, `/api/tamper`, `/api/power` and `/api/network` drive the physical world.
`panel.html`	The page. It refreshes every 400 ms and has four columns. **Cabinet screen:** the screen, login, dispense, deposit and the keypad. **Physical world (normal use):** door, compartment sensor and status. **Administrator / Technician:** reset, disarm, maintenance and active alarms, plus the off-normal events (tamper, power cut, network unplug). **Inventory:** stocked orders, orders waiting from the HMS, and recent activity.

`tests/`

`test_adc.py` has 21 tests in three groups:

- **`CabinetTests` (19):** the whole cabinet with a simulated clock and a fake HMS link, so the timing rules run instantly. Covers deposit then dispense, roles, the 10 s relock, the 60 s door limit (Invalid until the door closes), tamper while Invalid, the 3rd-failure warning and 5th-failure lockout, tamper, deposit blocks, the full cabinet, invalid order IDs, power loss, the session timeout, the HMS being unreachable, administrators being unable to open the cabinet, orders cancelled by the HMS, and the panel API.
- **`HmsCommTests` (1):** messages queue while offline and survive a restart.
- **`HmsRoundTripTests` (1):** a real cabinet against the real HMS over TCP, with the HMS clock 2 h off. Covers logins, order sync, stock, dispense, and checking the HMS log chain.

4. Operating modes



The tamper switch sends the cabinet to Locked Out from any mode, including Invalid, and sounds an alarm that only an administrator or technician can disarm. Invalid needs no reset: closing the door returns to Authorized with the same user logged in. The door is unlocked only in Dispense, Deposit and Door Open. After a transaction the user stays logged in (Authorized) until they log out or 5 minutes pass without activity.

5. Demo script for the TA

Open the HMS (http://127.0.0.1:8080) and the panel (http://127.0.0.1:8090) side by side. If the panel shows LOCKED OUT left over from an earlier run, reset it as tech1 first.

#	Do this	What to point out	Requirement
1	On the HMS, log in as doctor1 and order Insulin x5 and Morphine x2 for ADC-1	Both appear on the panel under "Waiting to be stocked (from HMS)"	FR-HMS-02
2	Panel: log in as pharmacist1, scan the Insulin order ID with destination `ADC-9`	Blocked: wrong ADC	FR-IVT-02
3	Dismiss, scan again with the destination left blank (it defaults to this cabinet), answer "No" to the quantity check	Blocked: quantity not confirmed	FR-IVT-04

#	Do this	What to point out	Requirement
4	Dismiss, scan again and answer "Yes". Open the door, press "Medication taken out / placed in", close the door	The order is Stocked in a compartment and the HMS shows Stocked. The pharmacist stays logged in	FR-IVT-01/05, FR-HMS-01/03/04
5	Log out. Log in as nurse1, type the Insulin order ID, press "Show order"	The order ID and medication are shown before dispensing	FR-DSP-02
6	Press "Dispense" and don't open the door for 10 s	The door locks again (relock countdown in Status); the nurse stays logged in	FR-SEC-10
7	Press "Show order" and "Dispense" again, open the door, press "Medication taken out / placed in", close the door	The HMS shows Fulfilled. Recent activity shows the time, user, order, medicine and quantity	FR-DSP-01, FR-HMS-01/03
8	Leave the cabinet alone while logged in	The session countdown runs down and any click restarts it. At 0 the user is logged out	FR-SEC-09
9	Logged out, press "Open door"	The door stays shut: "You must log in first"	FR-SEC-01
10	Fail the login 5 times (e.g. nurse1 with a wrong password)	Warning on the 3rd failure, LOCKED OUT on the 5th	FR-SEC-02/03, NFR-SEC-01
11	Try "Reset lockout" as doctor1, then as tech1	The doctor is refused and the technician resets it	FR-SEC-04
12	Press "Force open / damage (tamper)"	The alarm sounds (the buzzer blinks), the cabinet goes LOCKED OUT, and the HMS shows the alarm	FR-SEC-06/08
13	"Repair tamper", reset as tech1, then disarm as tech1 with a note	Reset alone does not silence the alarm. The disarm is logged with the time to disarm and notes	FR-SEC-07, FR-HMS-05
14	As pharmacist1, stock the Morphine order (press "Medication taken out / placed in") but leave the door open more than 60 s	Invalid state plus an alarm. Close the door: back to Authorized as pharmacist1, and the alarm stops by itself. Morphine stays Stocked	FR-SEC-05
15	Cut power, then restore power	Inventory, the event log and any lockout are kept	FR-HMS-06
16	Log in as nurse1 (logins need the HMS), then "Unplug network" and dispense the Morphine order	The HMS marks ADC-1 Warning, then Offline, once heartbeats stop. The panel's HMS pill shows the queued messages. "Plug network" sends them: ADC-1 goes Online and the order shows Fulfilled on the HMS	FR-HMS-07, NFR-HMS-01